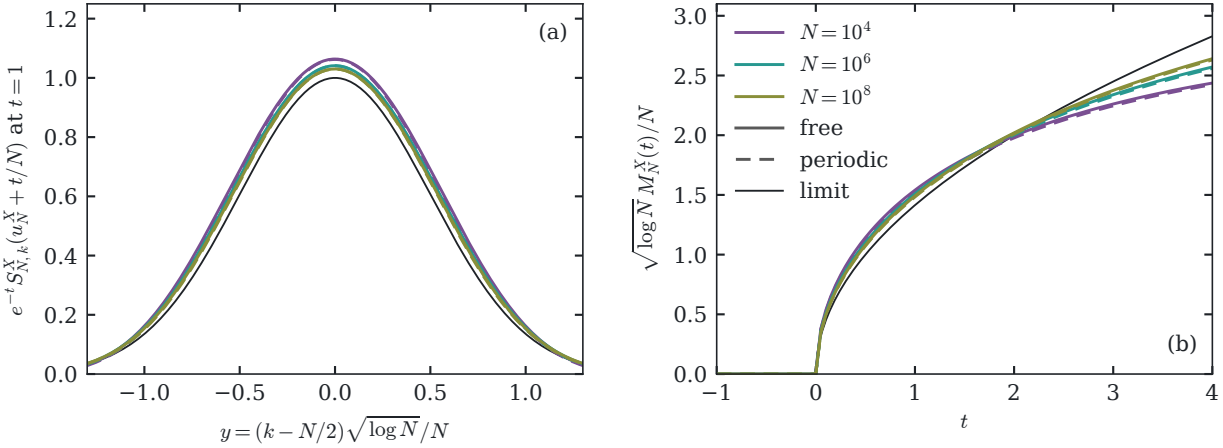
(a)
$e^{-t}S^X_{N,k}(u^X_N + t/N)$ at $t = 1$
$y = (k - N/2)\sqrt{\log N}/N$
(b)
$\sqrt{\log N}\, M^X_N(t)/N$
t
$N = 10^4$
$N = 10^6$
$N = 10^8$
free
periodic
limit